

ChemCraft: A Ludic Approach to Educational Game Design

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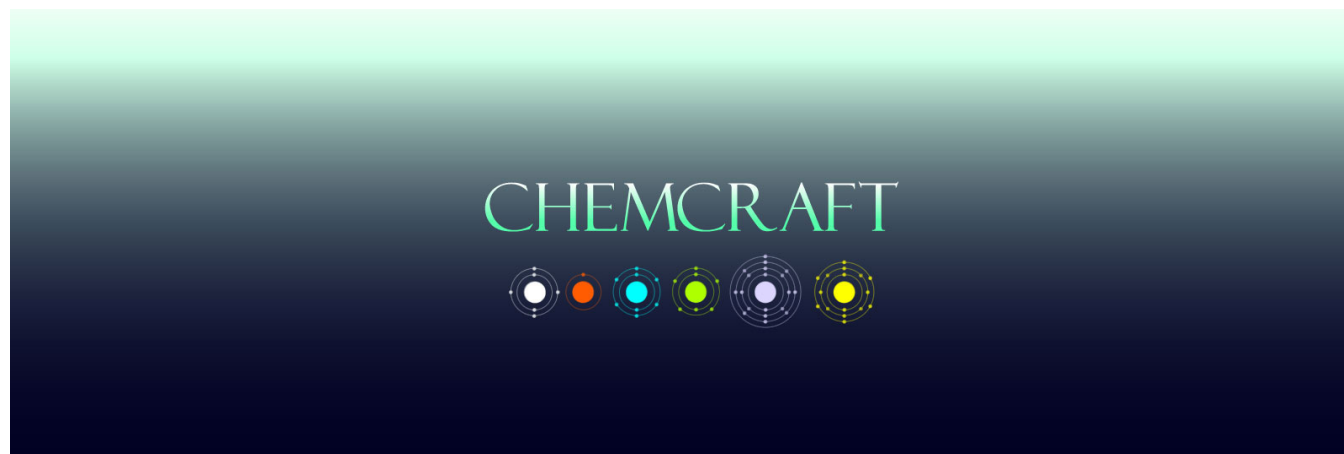


Figure 1: Title screen of *ChemCraft*, an adventure game for higher-level chemistry

ABSTRACT

We constantly both learn from and play with the world around us. We interact, we experiment, and we are curious; and through the spirit of Zimmerman’s ideas on ‘gaming literacy’, we can see the world as opportunities for play. This project thus presents *ChemCraft*, an adventure computer game based on chemistry. The game focuses on a ludic approach for educational game design, translating digital game rules from real chemistry rules: gameplay mechanics follow chemical reactions, game objects represent chemical compounds, and game object properties reference real chemistry data values. The game *ChemCraft* targets the niche field of games for higher education, specifically IB Higher Level Organic Chemistry, to investigate to what extent chemistry systems may be translated into game systems. The project asks not ‘What does a chemistry game look like’ but instead: ‘How can chemistry be a game?’

CCS CONCEPTS

- **Applied computing** → **Interactive learning environments;**
- **Human-centered computing** → **Graphical User Interfaces; User studies.**

KEYWORDS

Game Design, Educational Games, User Experience, Human-Computer Interaction

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1 INTRODUCTION

Educational games have a long but not particularly reputable history. Educational computer games have been developed as early as the 1950s [5] even before the commercialization of the gaming industry [8]. The problem frequently occurs when these educational games are released into the public, where the initial enthusiasm for educational games ends with disappointment [10]. The players are dissatisfied with either the educational value of the game or the element of fun in the game [3]. There persists a cycle of interest and disappointment in educational games, suggesting that the current models for educational game design need to be revisited. Well-known genres of educational games include serious games, persuasive games, and gamified education, but the project aims to develop an educational game through a more unconventional design approach — ‘gaming literacy’. “It [gaming literacy] asks, in other words, not *What does gaming look like?* but instead: *What does the world look like from the point of view of gaming?*” [20]. This paper thus presents the game *ChemCraft*, a 2D adventure game for higher level organic chemistry, to explore and revisit educational game design through ludic lens. Following the spirit of gaming

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literacy, the paper does not ask *What does a chemistry game look like?*, but instead: *How can higher level organic chemistry be a game?*

1.1 Educational Games: Research

A gap exists between the theory and practice of educational game development. The potential of educational games is discussed, educational game design models are proposed, and successful educational games are reviewed; however, there exists a relative lack of empirical evidence to substantiate the arguments proposed in this field. Nevertheless, the past 20 years have faced a significant resurgence in educational design research.

For instance, *What Video Games Have To Teach Us About Learning and Literacy* (2003) discusses how video games can help solve the educational dilemma of motivating and challenging students to learn [6]. *Moving Learning Games Forward* (2009) identifies obstacles for learning games, such as the negative sociocultural attitudes towards games and the perceived lack of seriousness of gaming amongst the educational community [12]. *Game-Based Learning* (2017) is a comprehensive guide on the spectrum of digital game-based learning [2]. A year ago, *The Effectiveness of Adaptive Versus Non-Adaptive Learning with Digital Educational Games* was published as an empirical study on whether adaptive games are superior to non-adaptive games, investigating how educational games ‘adapt’, or adjust, the difficulty level of its tasks to the performance level of the player [19]. Empirical investigations have started to become more common recently and offers abundant opportunity for further research.

Unfortunately, the challenge of designing good educational games still persists today and “the impact of educational games is lower than expected, especially the difficulty to adapt them to different educational contexts” [16]. Insufficient tried-and-tested game design strategies exist to adapt game models to diverse educational contexts, and more organic design approaches such as ‘gaming literacy’ are rarely explored.

1.2 Educational Games: Practice

According to a report at the Serious Play Conference, educational games are experiencing a boom phase – in 2018 alone 2.25 billion USD was invested in game-based learning companies [1]. This includes serious games, simulations, and gamification models. Nevertheless, the growing demand for educational games displays a comparatively limited uptake by consumers and game developers [18].

In a recent 2018 webinar, the game theorist Eric Klopfer suggests that educational games are received poorly by the public because they commit many of the same mistakes as traditional schooling: prescriptive learning, redundant questions, limited freedom to interact with the game world, etc. However, he articulates that such examples of educational games are misnomers for what actually are examples of edutainment, where “play is the reward for learning” [11] as opposed to the reverse where learning is a reward for play. The motivational value for the game is comparatively diminished.

Nevertheless, precedence of successful and novel educational games do exist, which should clearly be differentiated from edutainment. These include the online chemistry game based on protein-folding, *FoldIt*, a crowd-sourcing game where the player can contribute scientific research by designing proteins [7]. Another example is

ChemCaper, a role-playing game based on the IGCSE Chemistry syllabus [17], which features implicit learning mechanisms and attractive artwork. *Poptropica* is one of the few educational games that mentions a ‘gaming literacy’ to enjoy a storyline frequently grounded in factual history, focusing on contextual problem-solving skills and narrative engagement [13].

2 GAME DESIGN

ChemCraft is a ludic approach to educational game design. It follows the ludological definition of a game as “a rule-based formal system with variable and quantifiable outcomes” [9] and similarly applies this to Chemistry. Chemistry too is a rule-based formal system, in which different chemical reactions yield different values such as new compounds, heat, and other particles. Gaming literacy views the world as opportunities for play [20], and thus the game design of *ChemCraft* interprets real-world chemistry systems as opportunities for play.

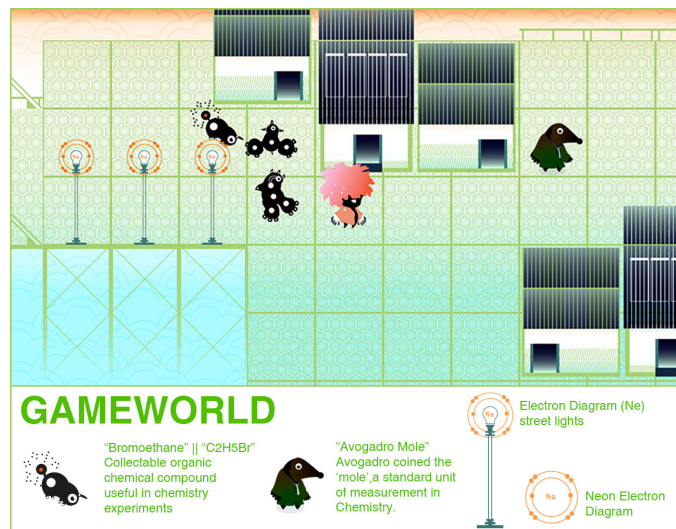
The game focuses on chemistry content offered at tertiary-level education – specifically, IB Higher Level Organic Chemistry [15]. To translate chemistry rules and knowledge into game mechanics and content, *ChemCraft* aims to adhere to the following principles whenever relevant.

- **Narrative elements** should incorporate chemistry knowledge and/or history.
- **Graphic representation** of game assets should provide meaningful visual cues into chemical structures.
- **Numerical values** in the game should reference chemistry data values.
- **Game rules and systems** should follow chemistry rules and systems.

2.1 Narrative Elements: Chemists, Chemical Compounds, and Chemistry Experiments

Chemistry education may be loosely divided into the following: chemists, chemical compounds, and chemistry experiments. *ChemCraft* translates these elements into game characters (chemists), collectables (chemical compounds), and battles (chemistry experiments).

- **Characters (Chemists)**: Game characters are modelled after chemists who coined key scientific concepts such as Avogadro’s Law, Markovnikov’s Rule, etc. As an adventure game, *ChemCraft* enables the player to converse, battle, and interact with different historical chemists and personified chemistry principles, ie. Avogadro’s Mole.
- **Collectables (Chemical Compounds)**: Collectables are modelled after organic chemical compounds, and their role is akin to Pokémon from the *Pokémon* franchise [14]. Chemical compounds in *ChemCraft* are sentient, are ‘raised’ by chemists, and have ‘battles’ with other chemical compounds via chemistry experiments.
- **Battles (Chemistry Experiments)**: Battles are modeled after chemistry experiments. *ChemCraft* applies a turn-based system for battling between a chemist’s team and the player’s team of chemical compounds. Battle moves are based on chemical reactions, ie. ‘combustion’, and thus a ‘battle’ embodies a chemical experiment.

Figure 2: Worldbuilding in *ChemCraft*

2.2 Graphic Representation: Chemical Structural Formulas

Game Representation	Structural Formula	Chemical Description
	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$	Methane: Alkane with 1 Carbon atom and 4 Hydrogen atoms. Empirical Formula: CH_4
	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{Br} \\ \\ \text{H} \end{array}$	Bromoethane: Halogenoalkane with 1 Carbon atom, 3 Hydrogen atoms, and a Bromide halogen atom. The halogen 'Br' is the Halo functional group. Empirical Formula: CH_3Br
	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{H} \end{array}$	Methanol: Alcohol with 1 Carbon atom, 3 Hydrogen atoms, and a Hydroxide ion. The hydroxide ion 'OH' is the functional group. Empirical Formula: CH_3OH
	$\begin{array}{c} \text{H} & & \text{H} \\ & \backslash & / \\ & \text{C}=\text{C} \\ & / & \backslash \\ \text{H} & & \text{H} \end{array}$	Ethene: Alkene with 2 Carbon atom and 4 Hydrogen atoms. Characteristic for the double bond between two carbon atoms, 'C=C', which is the functional group of this series. Empirical Formula: C_2H_4

HOMOLOGOUS SERIES
"group of chemical compounds with the same functional group with similar chemical properties which differ from each other by a common structural unit, such as $-\text{CH}_2-$ hydrocarbon."

Figure 3: Visual cues for different homologous series

A 'structural formula' is a graphical method of representing the location, type, and number of elements and bonds in a chemical compound [4]. Structural formulas provide insight into chemical properties and behaviour [15]. *ChemCraft* uses a system of visual cues to represent structural formulas as game art assets, aiming to communicate the following key organic chemistry concepts:

- **Structural Formula:** The elements and bonds in a chemical compound are represented graphically in *ChemCraft* through distinct symbols such as a large white circle for the carbon atom, a small white circle for the hydrogen atom, the OH hydroxide as a blue extension, and so on (see Fig.3). The

double bond for Alkenes is represented by two white lines connecting the two carbon atoms together. The size of the circles are size comparisons of the actual atoms themselves, i.e. reflective of how carbon is a bigger atom than hydrogen.

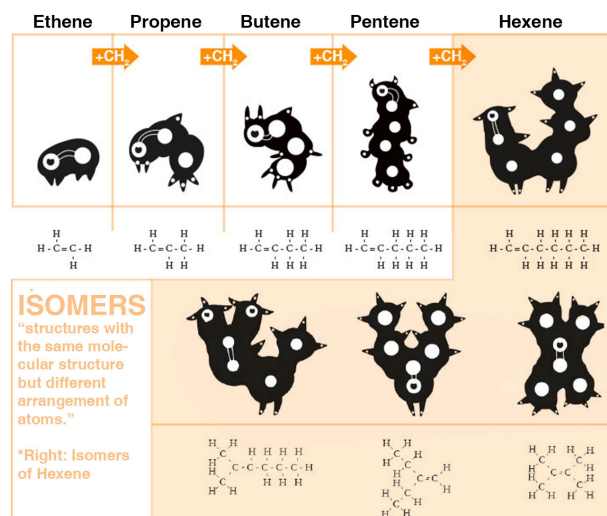


Figure 4: Structural Progression of Alkene Series

- **Homologous Series:** Akin to Pokémon 'types', homologous series are a chemistry term for groupings of organic compounds with similar properties. Homologous series are distinguished through their functional groups, which are graphically tagged in *ChemCraft* through the visual cues described in Fig.3. As a game property, functional groups determine the possible chemical reactions and thus the 'battle moves' of the compound.
- **Structural Isomers:** Isomers are compounds with the same molecular structure but different arrangement of atoms [4]. Fig.4 displays the Alkene homologous series, contrasting the progression of chemical compounds with increasing hydrocarbon (CH_2) chains with four isomers of a single chemical compound (hexene). As a game property, isomers determine how reactive the chemical is and thus the 'defense' rating of the compound, as the chemical is less likely to be affected by chemical reactions, or 'battle moves'.

2.3 Numerical Values: Chemistry Data Values

For as many relevant variables as possible, the game implements real chemistry values obtained from chemistry data bases such as the National Institute of Standards and Technology. These data values correspond to game status values, i.e. atomic mass (experience points) or combustion (damage value). Chemistry equations are programmed into the code itself as 'game rules'. Fig.5 displays how the 'health' value of a chemical compound is calculated via 'dissociation energy', the energy needed to break a compound into its constituent atoms [4] and the closest state to 'death' for a chemical.

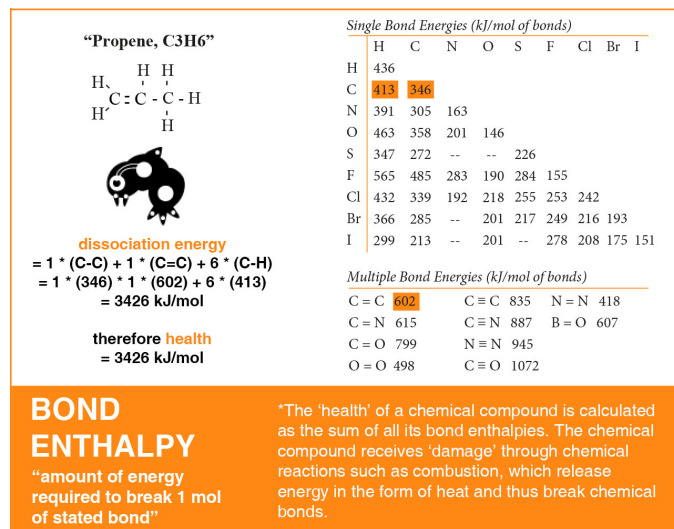


Figure 5: Game numerical values reflecting thermodynamic chemistry data

2.4 Game Rules: Chemical Reactions

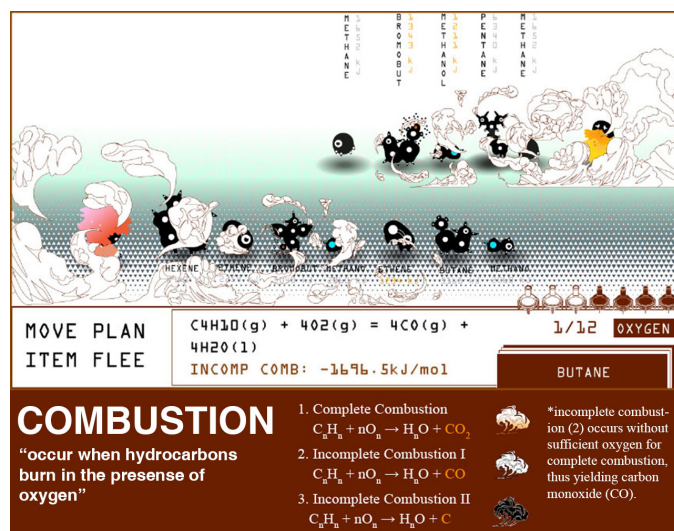


Figure 6: Game Rules defined by Thermodynamic Chemical Reactions

Following using real chemistry data values for game values, *Chem-Craft* uses chemical equations as its game rules. Two main types of chemical reactions are explored as game mechanics: combustion and synthetic reactions. Combustion is classified as a damage-dealing game action as they yield large amounts of heat via burning. Synthetic reactions are classified as a type of 'breeding' reaction because they yield new organic chemicals. Management of resources is also applied as a game mechanic because the player needs appropriate reactants in order to trigger a chemical reaction. For instance,

a successful combustion reaction requires sufficient oxygen in order to release heat. Fig.6 shows how a butane (C₄H₁₀) was only able to trigger one of three possible combustion reactions - incomplete combustion I, which yields carbon monoxide (CO) and lower damage value than complete combustion. The damage for combustion reactions is calculated within the code using a corresponding chemistry equation.

2.5 Technical Note

The game build targets Windows 10 PC and laptops, and the game was developed using Monogame 2013 in C#. The project development employed an iterative model as the game involved changing specifications. Requirements for the educational game were revisited with every new feature. The game was created by the author in its entirety — she was responsible for both programming the game engine in Visual Studio 2019 and producing all art assets in Adobe Illustrator and Photoshop.

3 PILOT STUDY

A pilot study was conducted to start investigating user feedback for this project. Five participants of over 18 years of age were invited to play *Chemcraft*. Participants were given the executable game to playtest remotely in their free time. Beyond general instructions on the controls for the game, there were no further instructions on how to play the game. Afterwards, participants were given three questionnaires: (1) an Intrinsic Motivation Inventory (IMI) with multiple-choice questions evaluating subjective experience; (2) three short essay questions regarding the enjoyment, challenge, and educational value of the game; (3) a six question pop-quiz of increasing difficulty on IB HL Organic Chemistry.

3.1 Discussion

The results of the IMI Likert analysis may be seen in Fig.7. Mean scores were calculated for each of the three IMI sections for a 7 point Likert scale. The participants believed that the game was quite fun (enjoyment = 6.03) and generally believed that they could find educational value in the game (usefulness = 5.46). The value for competence was close to the average of 4 (perceived competence = 4.76), meaning that the game was challenging but not overly so - as substantiated by an answer from the short essay section: "I found the battles challenging at first but quickly worked out how to set up combos, which was a rewarding experience".

Based on answers to the short-essay section, users who admitted to having prior knowledge in chemistry performed very well (6/6 correct) in the pop-quiz but responded that they only learned little. More informative, however, were the users who admitted to little or no knowledge in chemistry and performed comparatively poorly in the pop-quiz (2/6 correct). These users indicated that they learned about or even understood difficult chemistry terms such as "Free Radical Substitution" or "Enthalpy of Formation".

Regarding the enjoyment value of the game, users responded that they were intrigued by a spectrum of game elements: attractive graphics, exploration of the game world, and most notably, engaging gameplay. A user noted, "I found the application of the chemistry particularly interesting in the battle scenario."

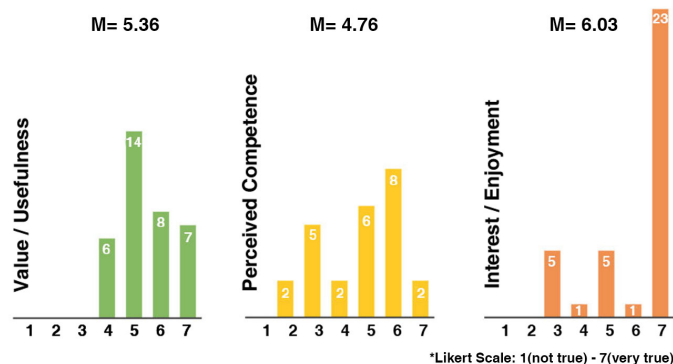


Figure 7: IMI Likert Charts: Educational Value/Usefulness, Perceived Competence, Interest/Enjoyment

3.2 Limitations

As the game *ChemCraft* is very content-specific, the game design decisions implemented are not easily adaptable for educational fields beyond chemistry. Additionally, the project would benefit from more extensive user studies. A pilot study has been conducted, but sample size is limited and the questionnaire could offer further empirical investigation.

4 CONCLUSION

Educational games serve as another lens to observe the reality around us, to interact with its processes at an advanced academic level, and to have fun learning through alternative means. This project presents *ChemCraft*, an educational game for IB Organic Chemistry, to provide a case study of game design guided by Zimmerman's ideas on gaming literacy. The project aims to create a game that is not prescriptive, not instructive, not restricting, but offers exploration, possibility, and experimentation. Through a ludic approach to game design, the project translates the rules of play from the rules of chemistry. Players of *ChemCraft* embark into a chemistry world, befriending sentient chemical compounds based on real chemical properties and battling historical chemists using real chemical reaction systems. Players may explore and learn at their own pace, with their own curiosity. In the beginning, this paper delineated the gap between the theory and practice of educational games. This project is a response to this issue, attempting to close the feedback cycle for designing, developing, and evaluating an educational game for higher education.

5 FUTURE WORK

The game has more technical issues to resolve: the robustness of the game engine needs to improve and the game would benefit from more content. A large amount of complex chemistry content is introduced in the game, and currently the game pacing is too fast to digest well. The game design framework *ChemCraft* currently proposes is applicable for many chemistry subjects, but mainly focuses on the Organic Chemistry module. However, Organic Chemistry is the tenth core module in the IB curriculum, and many of its

subjects build on the previous nine modules. Reflecting on the the players' experience of challenge from the user studies to expand the game content by including chemistry fundamentals would help ease players into learning about more difficult concepts and ideas. Moreover, a base comparison between *ChemCraft* and another educational game would be useful to position this work within the greater body of educational games design. Different players enjoy games differently, and *ChemCraft* presents just one of many ways to provoke learning and fun.

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